# Java Collections + Features

## Important 20 Coding Questions

**Instructions:** Write a complete Java program for each question. Use the given input and verify that your program produces the expected output.

### 1. Find Duplicate Elements

**Input:**

```
[10, 20, 30, 20, 40, 10, 50]
```

**Expected Output:**

```
Duplicate elements: [10, 20]
```

### 2. Find Frequency of Elements

**Input:**

```
[10, 20, 10, 30, 20, 10]
```

**Expected Output:**

```
10 → 3
20 → 2
30 → 1
```

### 3. Find Second Largest Number

**Input:**

```
[10, 50, 20, 80, 40, 70]
```

**Expected Output:**

```
Second largest: 70
```

### 4. Find Common Elements

**Input:**

```
List 1: [10, 20, 30, 40, 50]
List 2: [30, 40, 60, 70, 80]
```

**Expected Output:**

```
Common elements: [30, 40]
```

### 5. Find First Non-Repeated Element

**Input:**

```
[10, 20, 10, 30, 20, 40, 30]
```

**Expected Output:**

```
First non-repeated element: 40
```

### 6. Find First Repeated Element

**Input:**

```
[10, 20, 30, 20, 40, 10]
```

**Expected Output:**

```
First repeated element: 20
```

### 7. Sort Employees by Salary

**Input:**

```
101, Ravi, 50000
102, Suresh, 30000
103, Anil, 70000
104, Raj, 40000
```

**Expected Output:**

```
Suresh → 30000
Raj → 40000
Ravi → 50000
Anil → 70000
```

### 8. Sort Employees by Salary and Name

**Input:**

```
101, Ravi, 50000
102, Anil, 30000
103, Suresh, 50000
104, Raj, 30000
```

**Expected Output:**

```
Anil → 30000
Raj → 30000
Ravi → 50000
Suresh → 50000
```

### 9. Character Frequency Using HashMap

**Input:**

```
programming
```

**Expected Output:**

```
p → 1, r → 2, o → 1, g → 2,
a → 1, m → 2, i → 1, n → 1
```

### 10. Employee with Highest Salary Using HashMap

**Input:**

```
101 → Ravi → 50000
102 → Anil → 75000
103 → Suresh → 60000
104 → Raj → 45000
```

**Expected Output:**

```
Highest salary employee: Anil → 75000
```

# Java 8 Features — Stream API

### 11. Find Even Numbers Using Stream

**Input:**

```
[10, 15, 20, 25, 30, 35, 40]
```

**Expected Output:**

```
Even numbers: [10, 20, 30, 40]
```

### 12. Remove Duplicates Using Stream

**Input:**

```
[10, 20, 10, 30, 20, 40, 30]
```

**Expected Output:**

```
[10, 20, 30, 40]
```

### 13. Find Duplicate Elements Using Stream

**Input:**

```
[10, 20, 30, 20, 40, 10, 50, 30]
```

**Expected Output:**

```
Duplicate elements: [10, 20, 30]
```

### 14. Find Second Highest Using Stream

**Input:**

```
[10, 50, 20, 80, 40, 70]
```

**Expected Output:**

```
Second highest: 70
```

### 15. Find Second Lowest Using Stream

**Input:**

```
[50, 10, 30, 20, 40, 60]
```

**Expected Output:**

```
Second lowest: 20
```

### 16. Employee with Highest Salary Using Stream

**Input:**

```
101, Ravi, 50000
102, Anil, 80000
103, Suresh, 60000
104, Raj, 70000
```

**Expected Output:**

```
Highest salary employee: Anil → 80000
```

### 17. Employee with Second Highest Salary

**Input:**

```
101, Ravi, 50000
102, Anil, 80000
103, Suresh, 60000
104, Raj, 70000
```

**Expected Output:**

```
Second highest salary employee: Raj → 70000
```

### 18. Group Employees by Department

**Input:**

```
101, Ravi, 50000, IT
102, Anil, 60000, HR
103, Suresh, 70000, IT
104, Raj, 40000, HR
105, Kumar, 55000, Sales
```

**Expected Output:**

```
IT: Ravi, Suresh
HR: Anil, Raj
Sales: Kumar
```

### 19. Average Salary by Department

**Input:**

```
101, Ravi, 50000, IT
102, Anil, 60000, HR
103, Suresh, 70000, IT
104, Raj, 40000, HR
105, Kumar, 50000, Sales
```

**Expected Output:**

```
IT → 60000
HR → 50000
Sales → 50000
```

### 20. Highest-Paid Employee in Each Department

**Input:**

```
101, Ravi, 50000, IT
102, Anil, 60000, HR
103, Suresh, 70000, IT
104, Raj, 40000, HR
105, Kumar, 55000, Sales
106, Arjun, 65000, HR
```

**Expected Output:**

```
IT → Suresh → 70000
HR → Arjun → 65000
Sales → Kumar → 55000
```